Professional Diploma in

Project Management

Gantt Charts and Cost & Quality Management

Why Gantt?

The Gantt Chart was named after Henry L Gantt – an American engineer, management consultant and industrial advisor, who first devised it for production planning and resource loading.

Objectives

Lesson 4

- A closer look at Gantt charts
- Cost management
- Cost estimating techniques
- Quality management

The Gantt Chart

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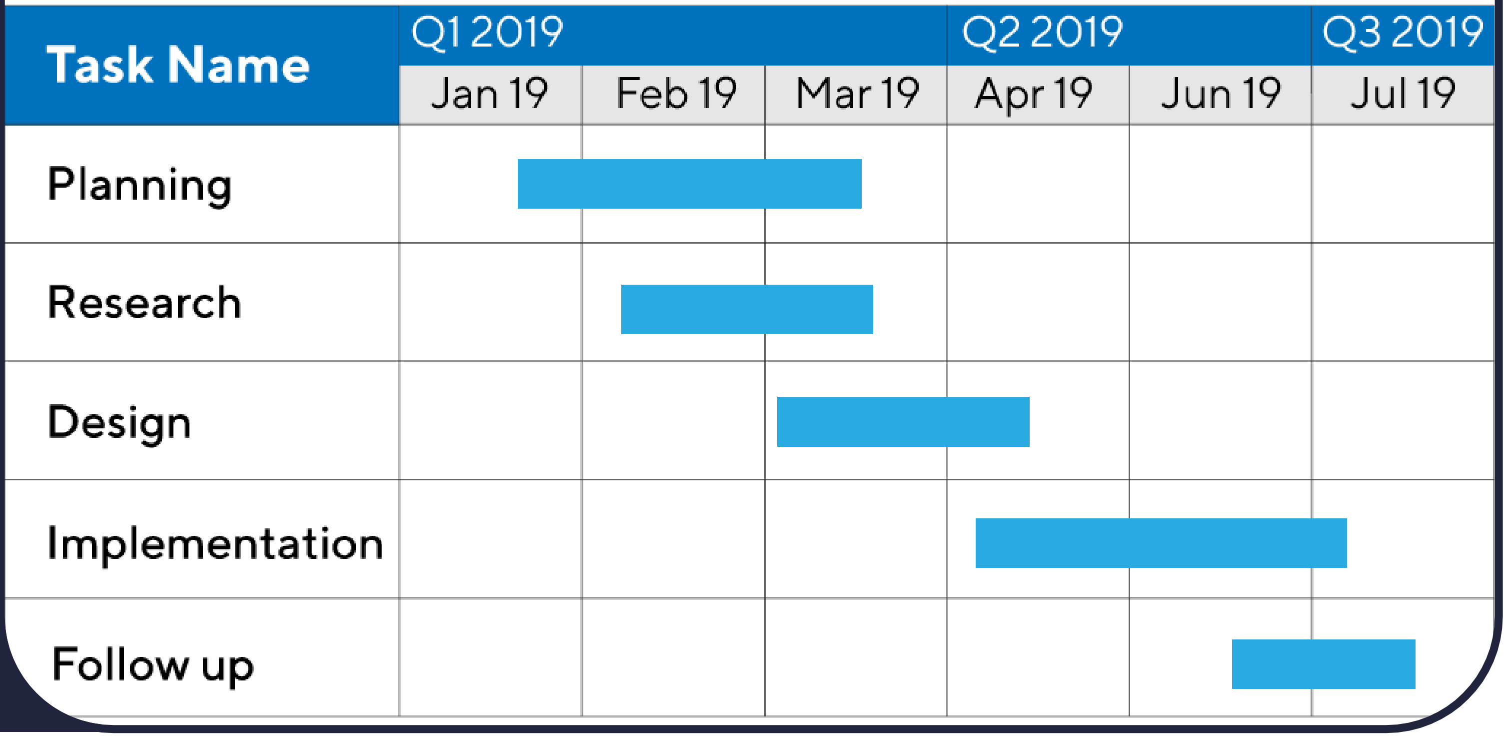
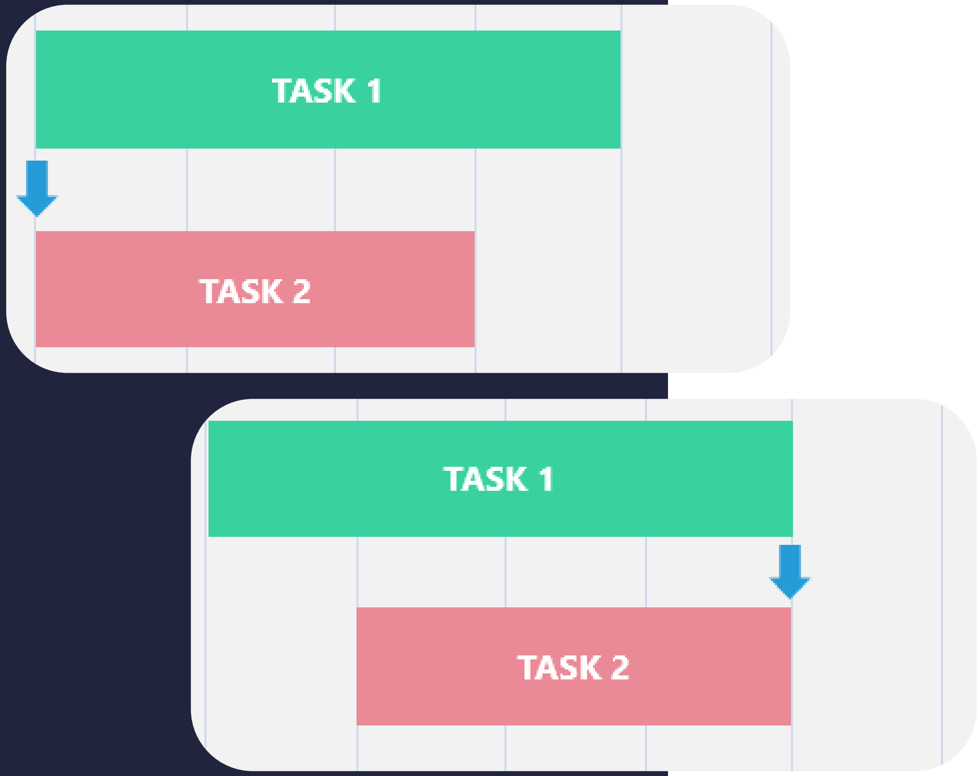
Gantt Chart



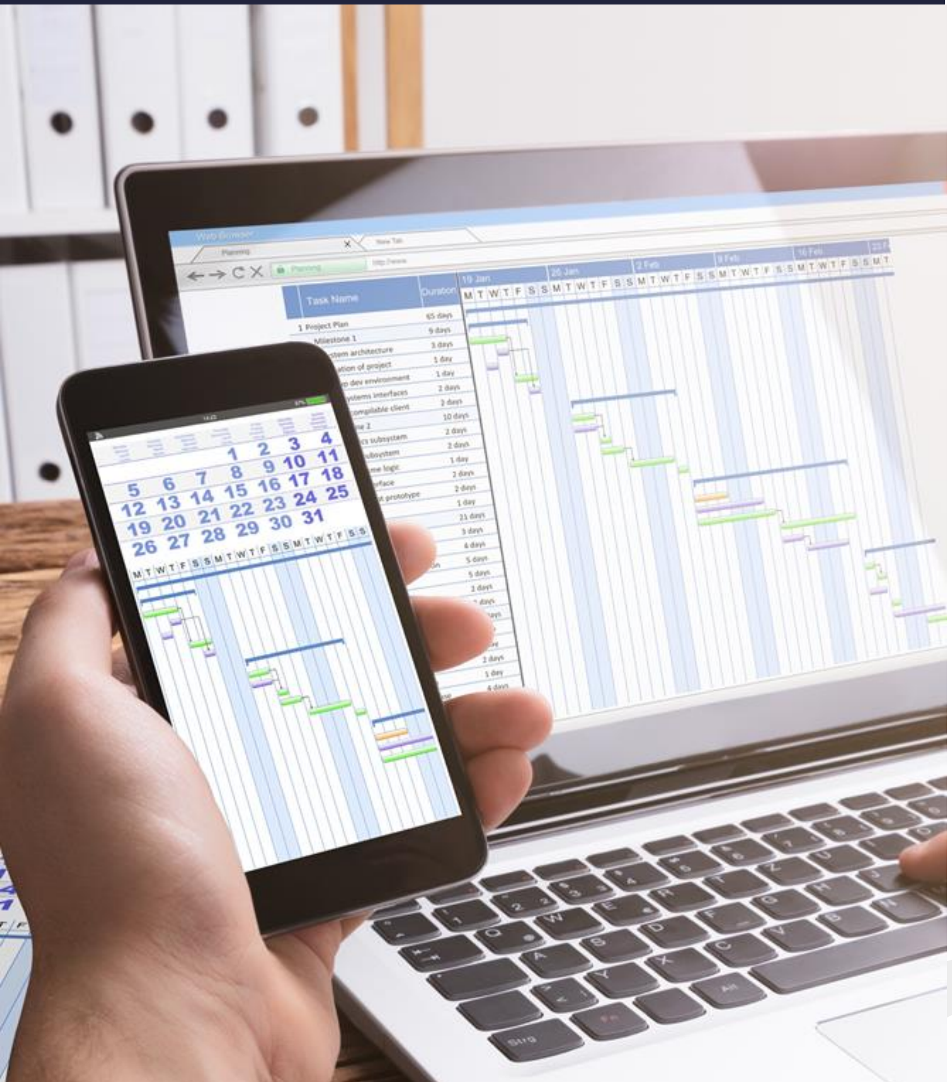
- Vertical axis: tasks to perform and scheduling a project's tasks
- Horizontal axis: time intervals as bars

Dependencies



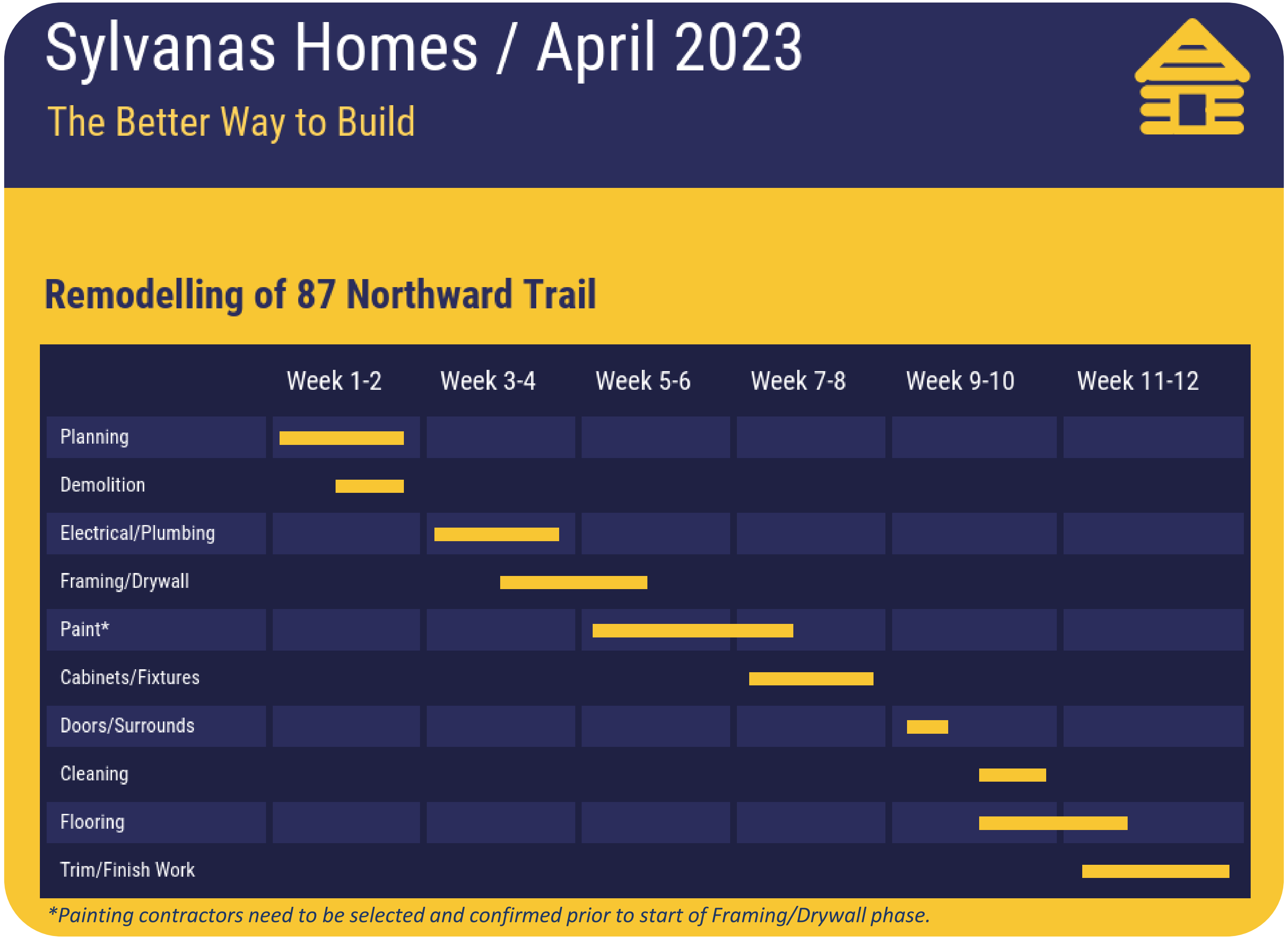
- Define relationships between tasks
- Visualise the sequence in which they must be completed
- Predecessor and successor then relationship between them

Benefits of a Gantt Chart?



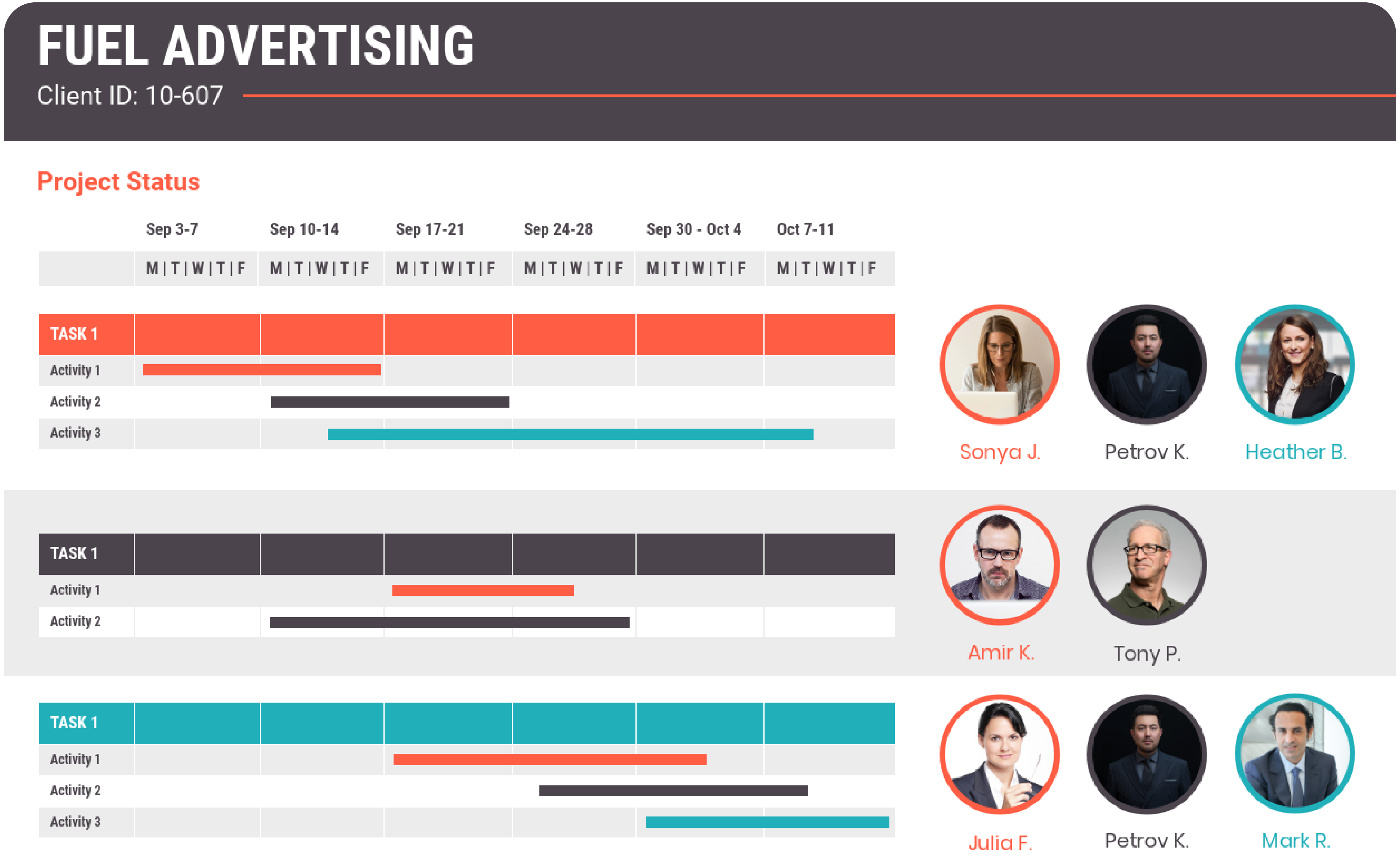
- Improves team communication and productivity
- Helps with resource planning and allocation
- Summarises complex data to enable easy monitoring and tracking
- Allows for project adjustments
- Identifies risks and issues to be resolved

Gantt chart examples



Gantt chart example

Gantt chart example



Cost management

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Cost management

- Process of estimating and controlling expenses
- Predict future expenses Project costs calculated during the planning phase
- Expenses are documented and tracked
- Upon project completion, predicted costs vs actual costs are compared

Four main phases of cost management

Determine budget

- Project budgets allow project managers to make smart decision regarding cost, scope, & time (triple constraint)
- Budget takes into account all monetary resources needed to achieve project goals

To establish a budget you will need:

- Risk registers
- Cost management plan
- Project schedule
- Resource calendars
- Scope baseline
- Cost estimates

Control costs

- Monitor all costs in real-time to ensure they stay aligned with the budget
- Chart out task and team-based costs
- Track productivity to estimate billable hours
- Set task-based and hourly rates as required

Control costs

- Allows opportunities to identify ways of reducing the expenses within the project which can turn increase overall profits
- Involves accounting for financial risks that could arise
- If costs not controlled, project runs the risk of the project going over budget and not aligning with scope

5 tips for successful cost control:

1. Capture the entire scope in your WBS
2. Insist on input and collaboration from outside stakeholders
3. Determine cost categories
4. Develop trust within your team
5. Take action immediately

Different cost categories

Direct cost

Directly linked to your specific project output

Example: project software

Indirect cost

Not linked directly to your project

Example: office space rental

Fixed cost

Once-off expenses

Example: buying a company vehicle

Variable cost

Costs that are subject to change

Example: petrol/gas costs

Sunk cost

Payment has already been made

Example: market research before a project

Other useful ‘cost’ terms

Unburdened cost

Cost related to labour

Example:
base salaries

Burdened cost

Employee benefits

Examples: health insurance, car allowances

Material cost

Physical equipment

Example:
laptop

Ancillary cost

Any additional cost

Examples:
training,
travel

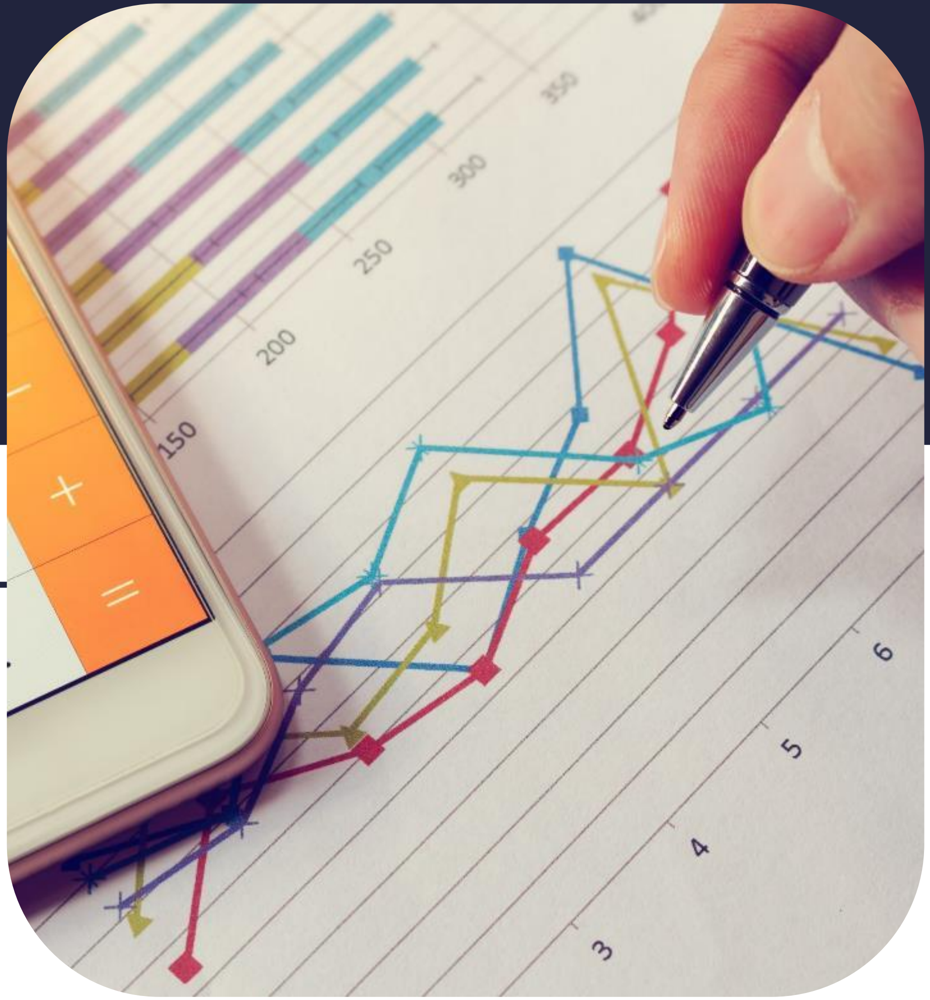
Estimating costs

The process of forecasting to achieve the project scope

The forecast or estimate can often influence the decision to continue the project (RAG)

Estimates don’t need to be perfect

Costs can be broken down into
:
- Direct costs
- Indirect costs



Cost estimating techniques

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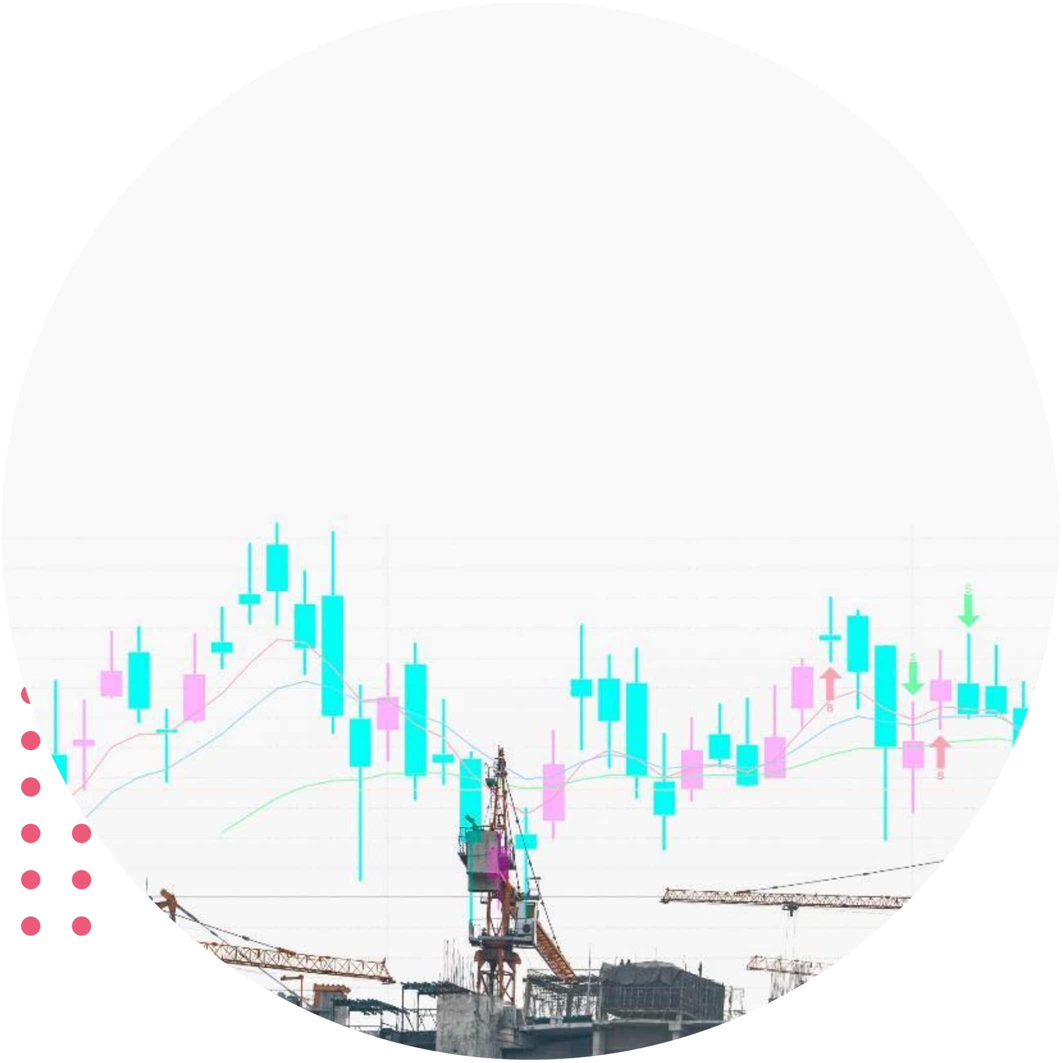


Parametric estimation technique

- Identify unit cost/duration
- Estimate number of units required
- High accuracy
- Time efficient

Parametric estimation example

- Room to be painted:
  2,5 m high (h) and 10 m wide (w)

- Historical data demonstrates:
  - 20 minutes to paint 1 m^2
  - 1 litre of paint required for 1 m^2

Parametric estimation example

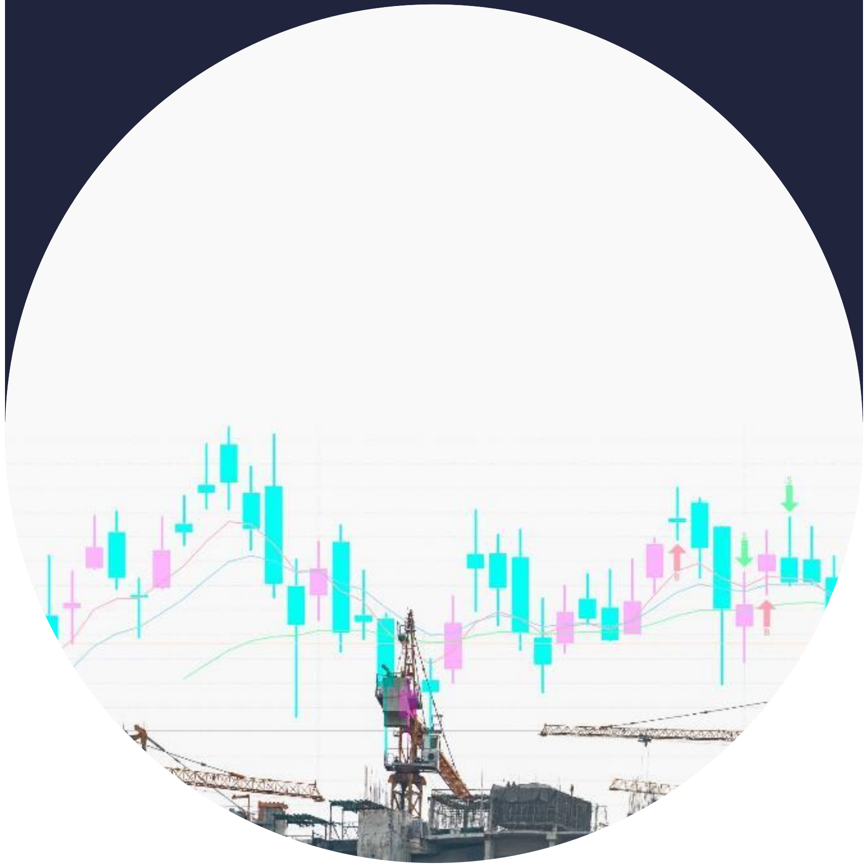
- Total square meterage

  = 2,5 (h) x 10 (w) = 25 m^2

- 25 m^2 x 20 min = 500 min/60 min

  = 8 H 33 min of labour

- 25 m^2 x 1 L = 25 litres of paint



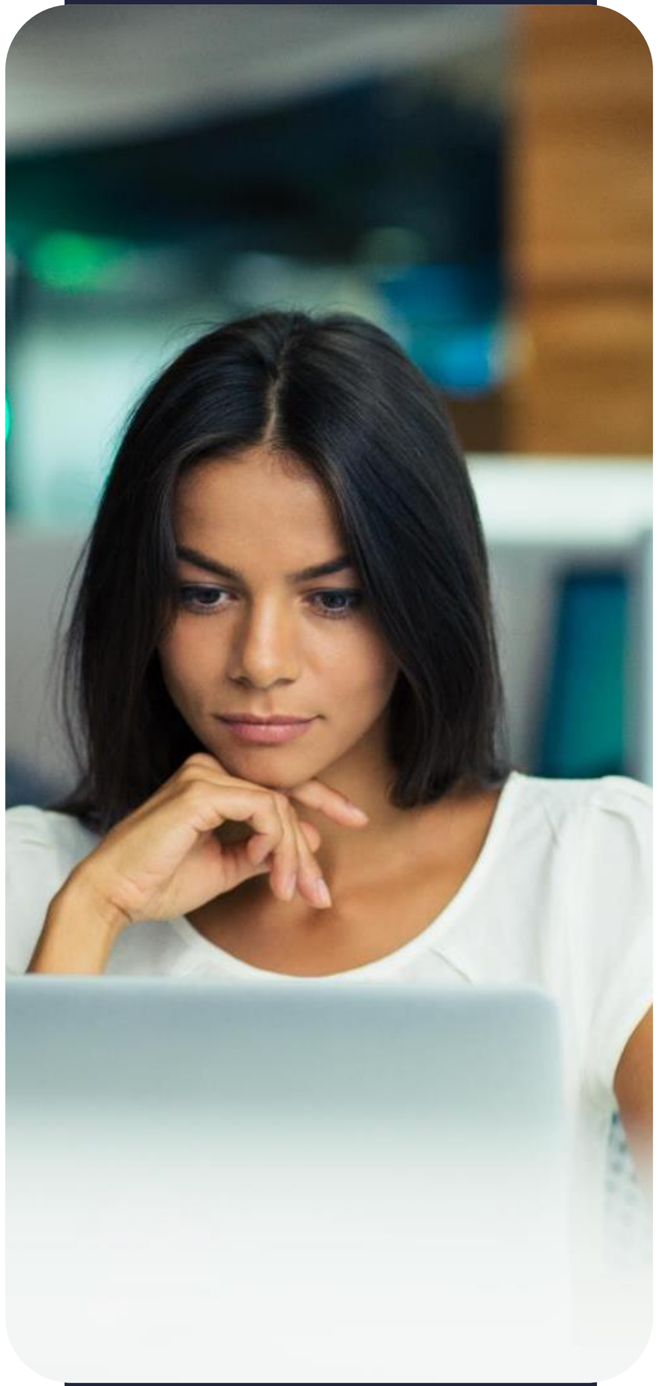
Analogous estimation technique

- Uses similar past project to estimate current project
- Easy to use
- Option when there is limited information available

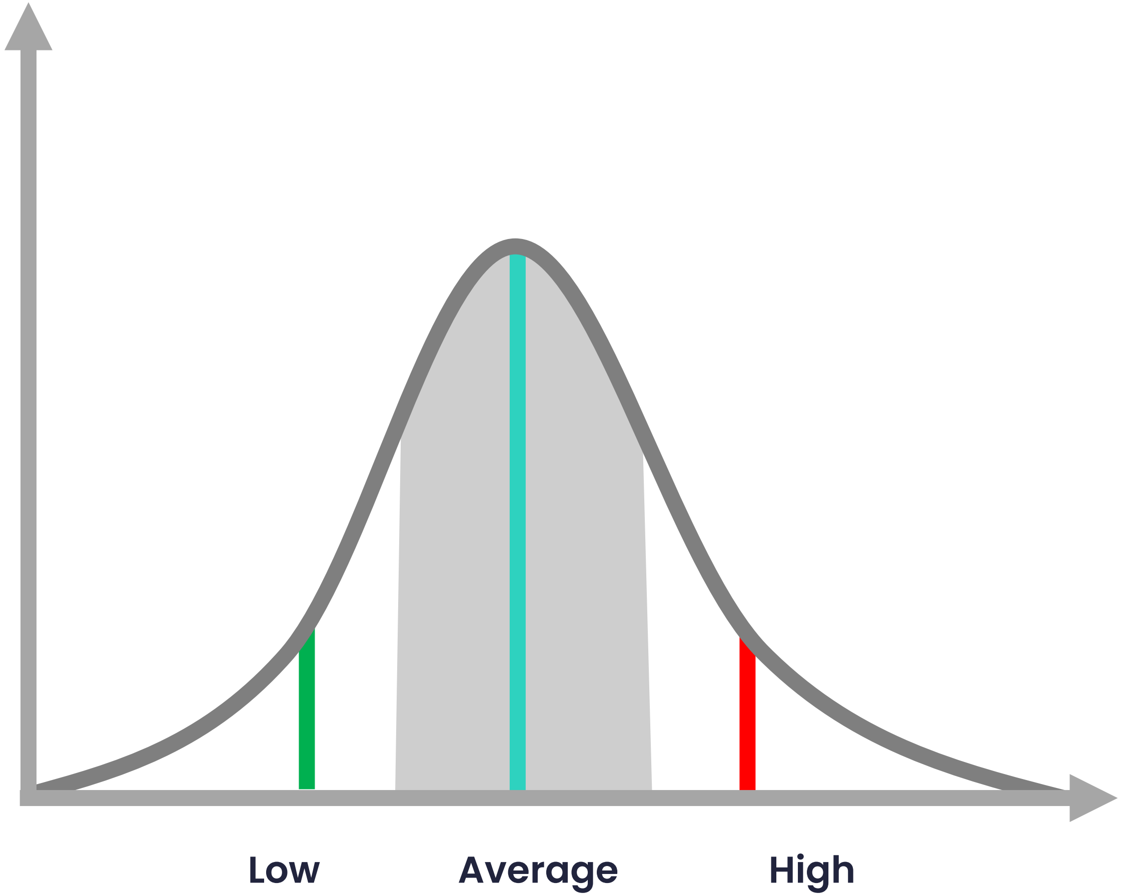
Analogous estimation example

Tina's advertising agency is bidding on two different commercial campaigns. One is a digital campaign for a new resort in town, and the other is for a dating website.

The agency has done a digital advert campaign for a new hotel in the past, so they collect data from that campaign. The agency is better equipped to make an initial estimation for this project based on analogous estimating than the dating website campaign for which they do not have relevant past data available.



Bell curve technique



Quality management

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Quality management is the process for ensuring that all the activities necessary to design, plan, and implement a project are effective and efficient.

What is quality?

What the customer or stakeholder needs from the project deliverables

Key focus of quality management

Customer satisfaction

- When planning, understand the customer requirements
- Allows you to deliver an end result that satisfies
- Their satisfaction facilitates a positive and trust-based relationship that could lead to future projects

How many customers complain?

Only 4% of customers complain; the rest keep silent and simply go elsewhere!

Tips for satisfied customers

- **Keep customers informed**
  - Progress checks
  - Weekly meetings
  - Return phone calls/emails

Tips for satisfied customers

- **Under promise & over deliver**
- **Take ownership when things go wrong**
- **Don’t disregard what the customer is saying because ‘you know better’**

Quality plan

- Alignment with customer requirements
- Alignment with internal & external stakeholders
- Problems within the project
- Overall conformance to the project plan

The Quality Management Plan (QMP)

- Quality objectives
- Key project deliverables and processes
- Quality standards
- Quality control & assurance activities
- Quality roles & responsibilities
- Quality tools
- Plan for reporting quality control and assurance problems

The QMP describes how quality will be managed throughout the lifecycle of the project.

Quality assurance

Process oriented – ensures that desired quality requirements are met

Evaluate processes & procedures

Helps prevent defects or deviations from the requirements

Quality control

Product oriented – focuses directly on the deliverable

Specifications at multiple stages of the project

Present throughout the lifecycle of the project so that adjustments can be made as needed

User acceptance testing

- Used in the IT industry
- Designated testers are challenged with breaking or finding gaps in system functionalities or design
- Bugs/defects are captured & developers/designers need to rectify the problem
- Their solution is then tested again until it meets user requirements

Where are we?

- Project Management lifecycle
- Initiation Phase
- Planning
  - Cost and Quality Management

Where are we?

- Project Management lifecycle
- Initiation Phase
- Planning
  - Cost and Quality Management
  - Control, Risk, Procurement and Acceptance Management
- Project execution and leadership
- Monitoring and controlling
- Project closure